Homework 1

Collaboration with other students are encouraged; remember to name anyone you collaborate with and to write up your final solution by yourself.

1. Show that in any base $b \geq 2$, the sum of any three single-digit numbers is at most two digits long.

2. Prove that if a natural number $N > 1$ has no prime divisors less than or equal to $\sqrt{N}$, then N is prime.

3. Consider the problem of computing $N! = 1 \cdot 2 \cdot 3 \cdots N$. If N is an n-bit number, how many bits long is $N!$, approximately (in $\Theta(\cdot)$ form)?

   Hint: Use Stirling's approximation

4. Show that

$$\sum_{i=1}^{n} \frac{1}{i} = \Theta(\log(n))$$

   Hint: To show an upper bound, decrease each denominator to the next power of 2. For a lower bound, increase each denominator to the next power of 2.

5. Let p be the polynomial defined by $p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$. Suppose we want to evaluate $p(x)$ for a given real number x.

Algorithm: Polynomial Evaluation

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EvaluatePolynomial(a_0, a_1, ..., a_n, x):
    z := a_n
    i := n
    while i > 0 do
        i := i − 1
        z := z · x + a_i
    return z
```

   Show that $z = \sum_{j=i}^{n} a_j x^{j-i}$ is an invariant that the algorithm maintains.

6. Describe a polynomial-time algorithm for computing $a^k \pmod{p}$, given a, k, p. Show that the algorithm indeed runs in polynomial time.